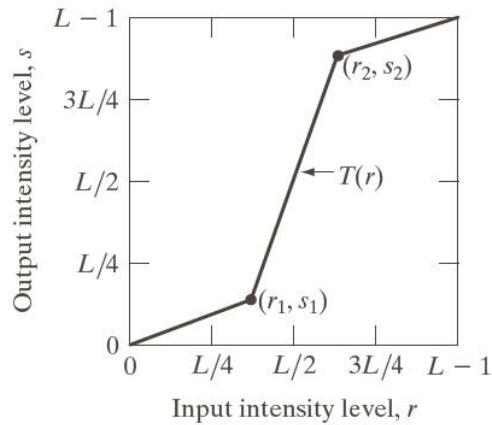


Today's Calendar

Intensity Transformation

Histogram processing

Piecewise-Linear Transformation Functions: Contrast Stretching



a b
c d

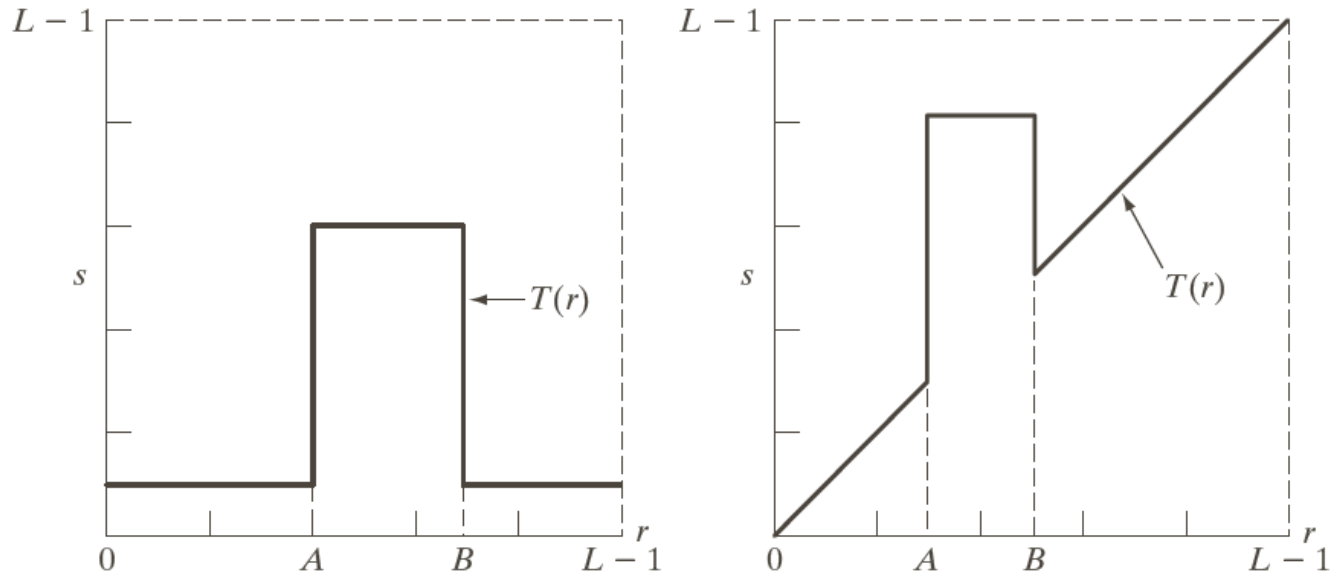
FIGURE 3.10

Contrast stretching.
(a) Form of transformation function. (b) A low-contrast image. (c) Result of contrast stretching. (d) Result of thresholding. (Original image courtesy of Dr. Roger Heady, Research School of Biological Sciences, Australian National University, Canberra, Australia.)

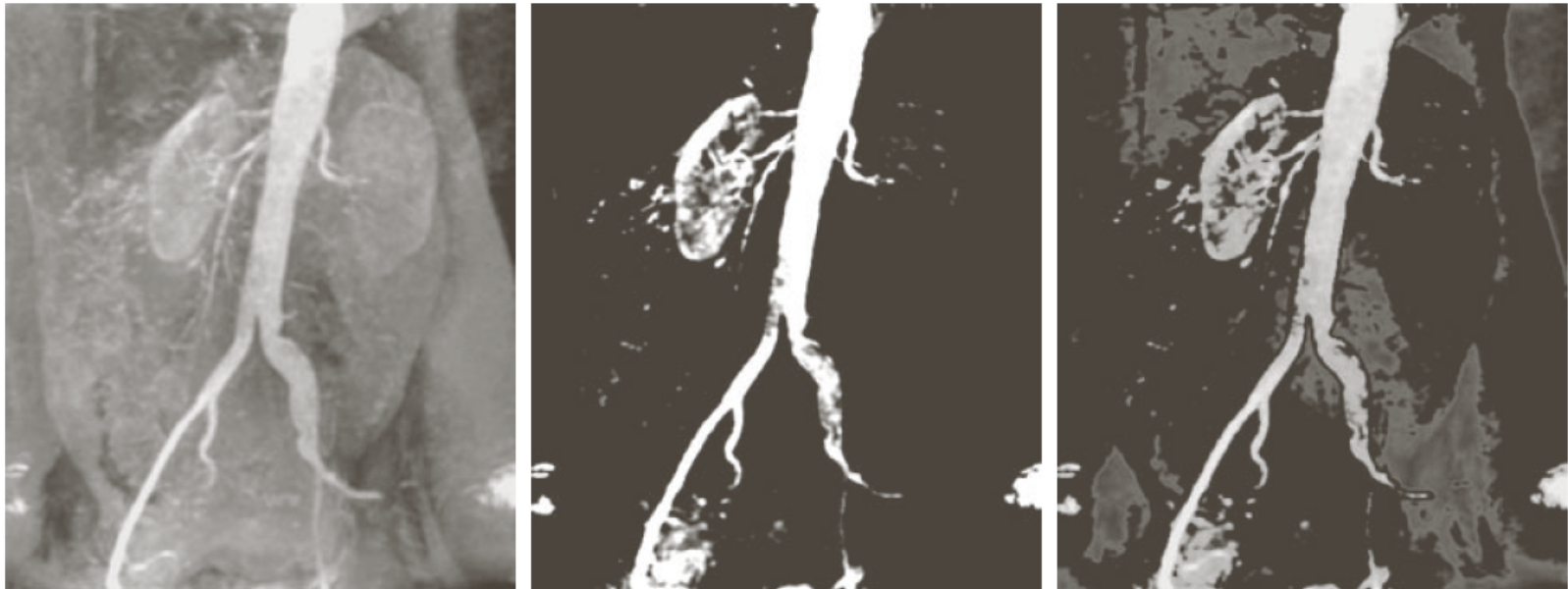
Piecewise-Linear Transformation Functions: Intensity-Level Slicing

a b

FIGURE 3.11 (a) This transformation highlights intensity range $[A, B]$ and reduces all other intensities to a lower level. (b) This transformation highlights range $[A, B]$ and preserves all other intensity levels.



An Example of Intensity-Level Slicing



a b c

FIGURE 3.12 (a) Aortic angiogram. (b) Result of using a slicing transformation of the type illustrated in Fig. 3.11(a), with the range of intensities of interest selected in the upper end of the gray scale. (c) Result of using the transformation in Fig. 3.11(b), with the selected area set to black, so that grays in the area of the blood vessels and kidneys were preserved. (Original image courtesy of Dr. Thomas R. Gest, University of Michigan Medical School.)

Piecewise-Linear Transformation Functions: Bit-Plane Slicing

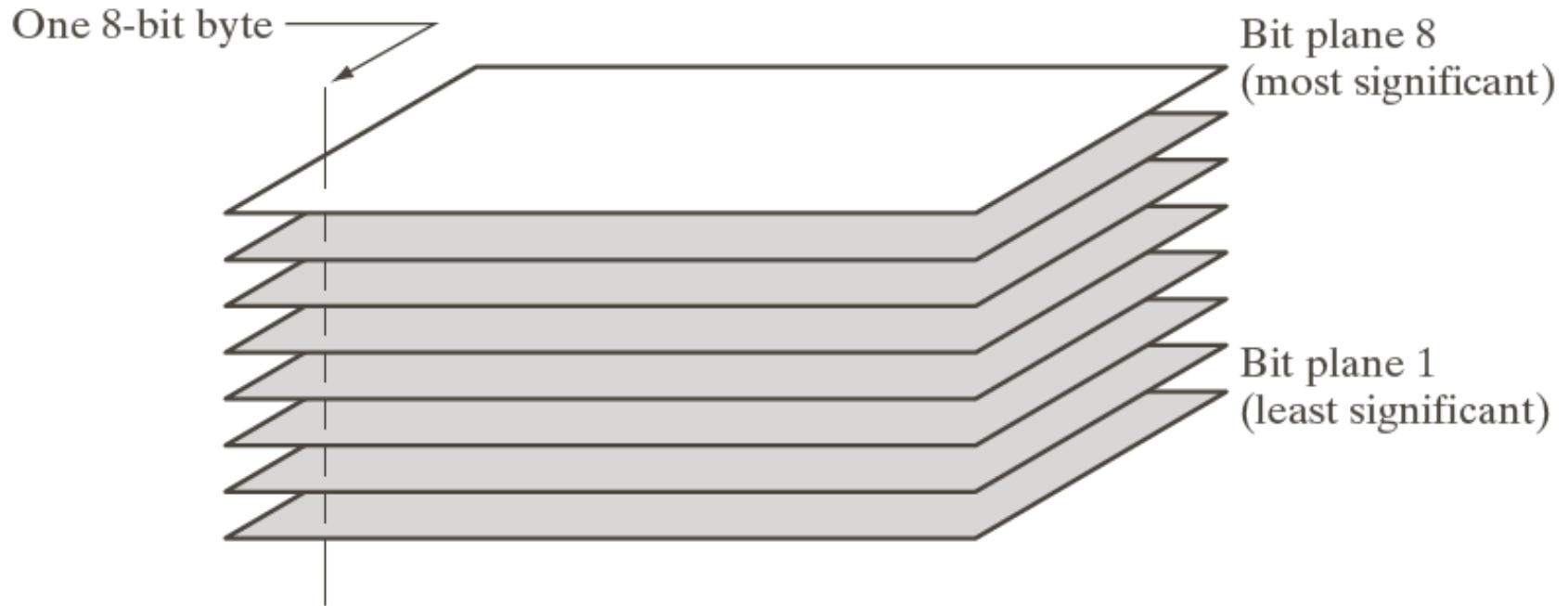


FIGURE 3.13
Bit-plane
representation of
an 8-bit image.

An Example

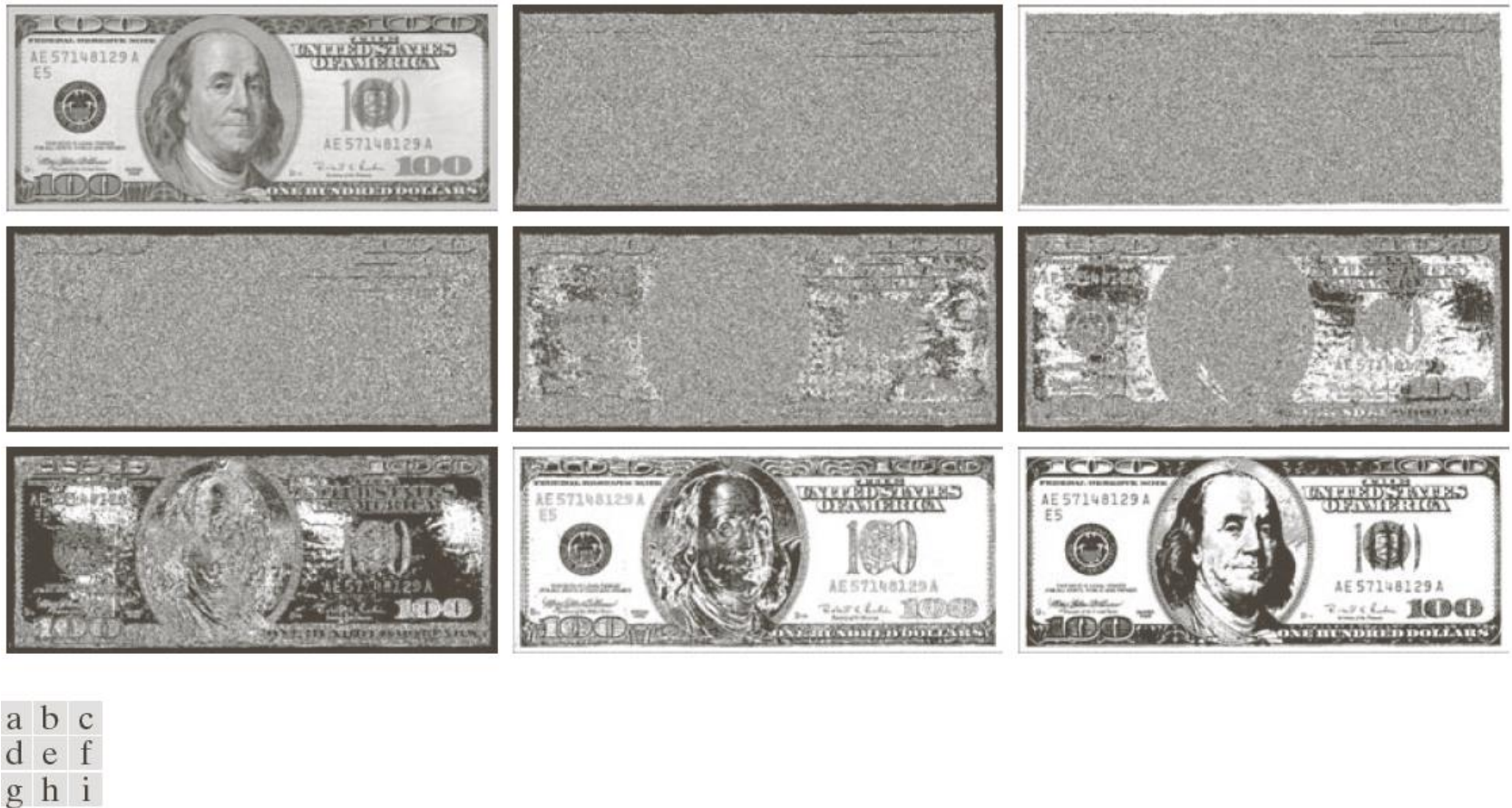


FIGURE 3.14 (a) An 8-bit gray-scale image of size 500×1192 pixels. (b) through (i) Bit planes 1 through 8, with bit plane 1 corresponding to the least significant bit. Each bit plane is a binary image.

Use for Image Compression

original



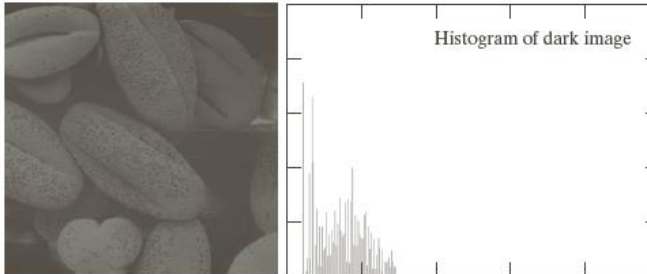
a b c

FIGURE 3.15 Images reconstructed using (a) bit planes 8 and 7; (b) bit planes 8, 7, and 6; and (c) bit planes 8, 7, 6, and 5. Compare (c) with Fig. 3.14(a).

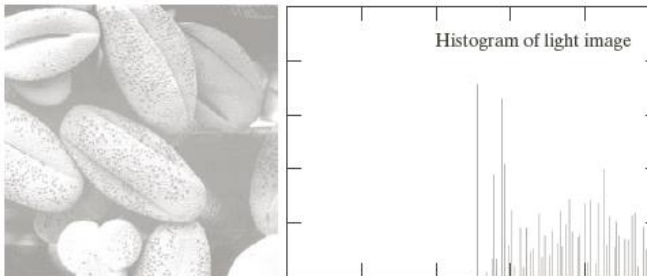
Less bit planes are sufficient to obtain an acceptable details, while require half of the storage

Histogram Processing

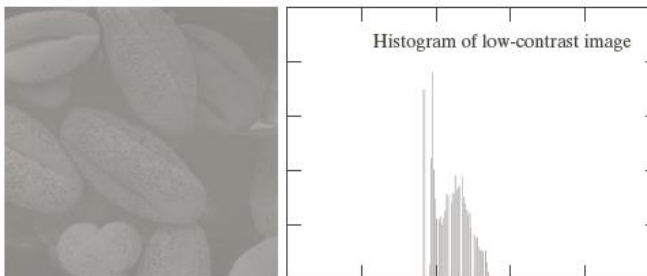
Dark:



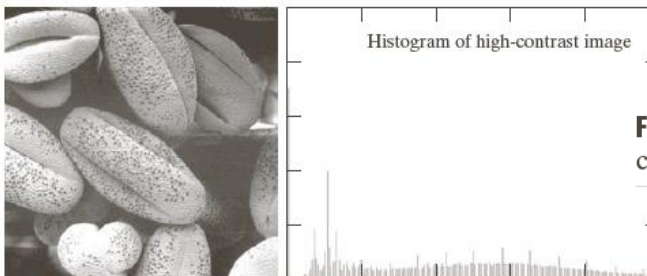
Light:



Low
contrast:



High
contrast:



Histogram

$$h(r_k) = n_k$$

Normalized histogram

$$p(r_k) = n_k / MN$$

$$\sum_{k=0}^{255} p(r_k) = 1$$

FIGURE 3.16 Four basic image types: dark, light, low contrast, high contrast, and their corresponding histograms.

Transformation Function

$$s = T(r) \quad 0 \leq r \leq L-1$$

A valid transformation function must satisfy two conditions:

(a) $T(r)$ is monotonically increasing, i.e., $T(r_1) \geq T(r_2)$ if $r_1 > r_2$

(b) $0 \leq T(r) \leq L-1$ The same range as input

(a') $T(r)$ is strictly monotonic : one - to - one mapping $r = T'(s)$

Intensity Transformation Function

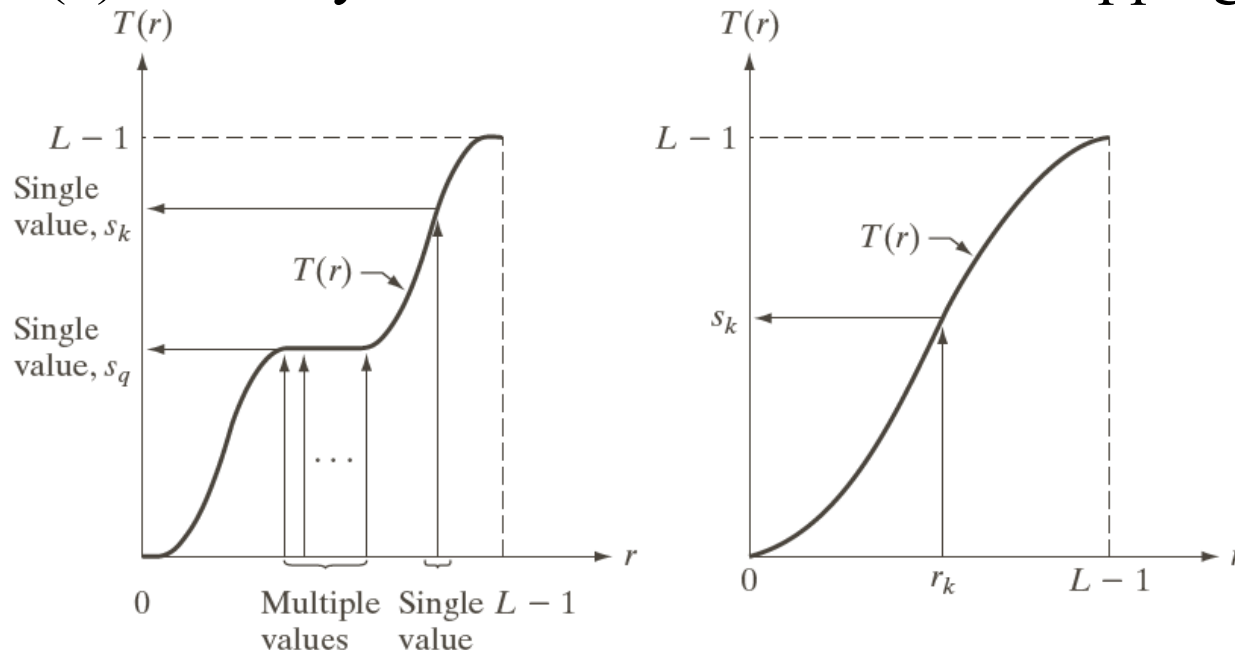
$$s = T(r) \quad 0 \leq r \leq L-1$$

A valid transformation function must satisfy two conditions:

(a) $T(r)$ is monotonically increasing, i.e., $T(r_1) \geq T(r_2)$ if $r_1 > r_2$

(b) $0 \leq T(r) \leq L-1$

(a') $T(r)$ is strictly monotonic : one - to - one mapping $r = T'(s)$

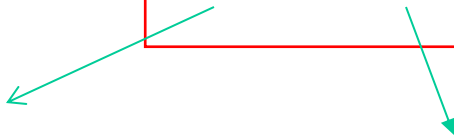


a b

FIGURE 3.17
(a) Monotonically increasing function, showing how multiple values can map to a single value.
(b) Strictly monotonically increasing function. This is a one-to-one mapping, both ways.

Histogram Processing

If $T(r)$ is continuous and differentiable over the range of r , then

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|$$


Probability density function of intensity value

Histogram Equalization

A special transformation function

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$


Cumulative distribution function of r

Is it a valid transformation function?

Yes.

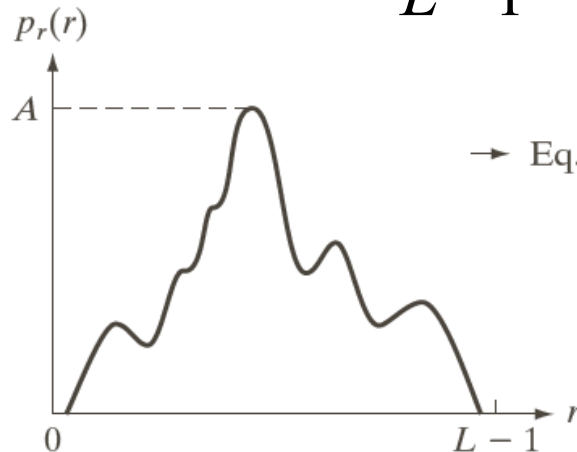
(a) $T(r)$ is monotonically increasing, i.e., $T(r_1) \geq T(r_2)$ if $r_1 > r_2$

(b) $0 \leq T(r) \leq L-1$

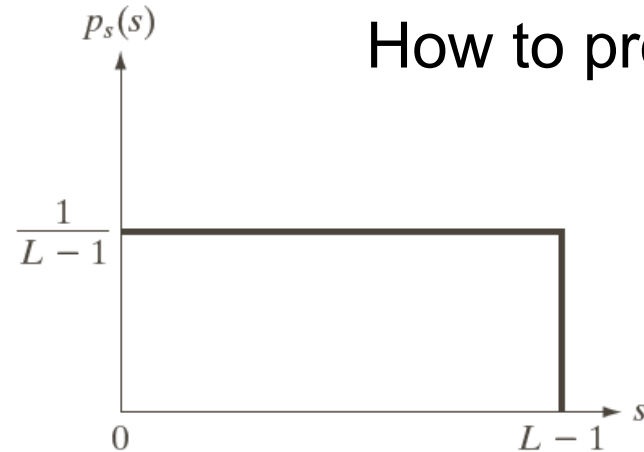
Histogram Equalization

$$s = T(r) = (L - 1) \int_0^r p_r(w) dw$$

→ $p_s(s) = \frac{1}{L-1}$



→ Eq. (3.3-4) →



How to prove it?

a b

FIGURE 3.18 (a) An arbitrary PDF. (b) Result of applying the transformation in Eq. (3.3-4) to all intensity levels, r . The resulting intensities, s , have a uniform PDF, independently of the form of the PDF of the r 's.

Histogram Equalization

$$p_s(s) = p_r(r) \left| \frac{dr}{ds} \right|, \quad s = T(r) = (L-1) \int_0^r p_r(w) dw$$

$$\longrightarrow p_s(s) = \frac{1}{L-1}$$

An Example

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & 0 \leq r \leq (L-1) \\ 0 & \textit{otherwise} \end{cases}$$

$$s = (L-1) \int_0^r p_r(w) dw = \dots = \frac{r^2}{L-1}$$

Histogram Equalization – Discrete Case

$$p_r(r_k) = n_k / MN, k = 0, 1, 2, \dots, L-1$$

$$s_k = T(r_k) = (L-1) \sum_{j=0}^k p_r(r_j) = \frac{L-1}{MN} \sum_{j=0}^k n_j$$

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1
Intensity
distribution and
histogram values
for a 3-bit,
 64×64 digital
image.

$$s_0 = T(r_0) = (L - 1) \sum_{j=0}^0 p_r(r_j) = (8 - 1) * 0.19 = 1.33 \rightarrow 1$$

$\vdots$

$$s_7 = T(r_7) = (L - 1) \sum_{j=0}^7 p_r(r_j) = (8 - 1) * 1 = 7$$

Histogram Equalization – Discrete Case

r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

TABLE 3.1

Intensity distribution and histogram values for a 3-bit, 64×64 digital image.

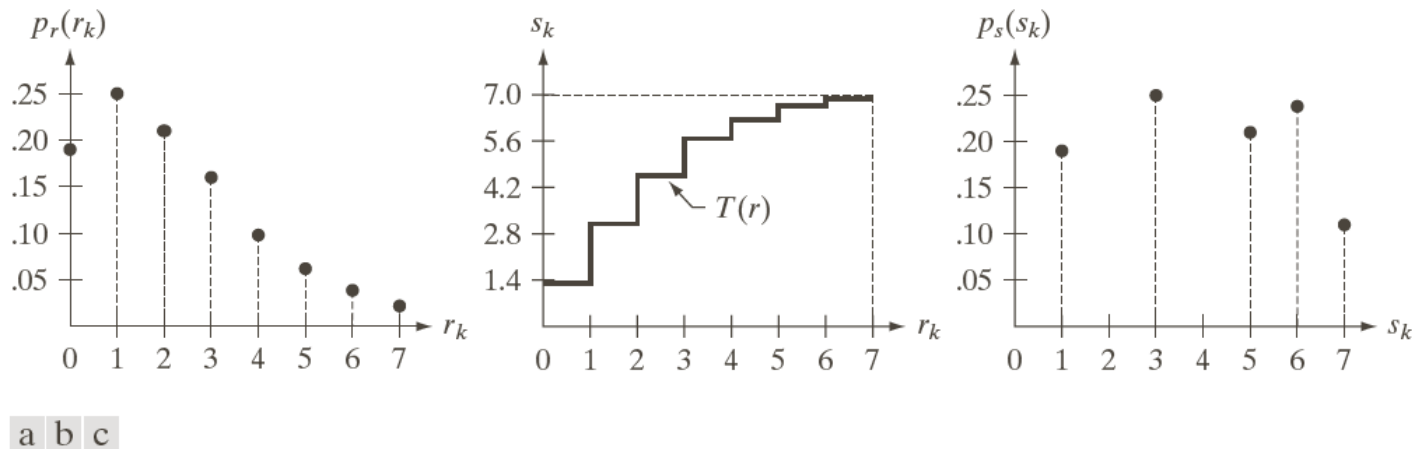


FIGURE 3.19 Illustration of histogram equalization of a 3-bit (8 intensity levels) image. (a) Original histogram. (b) Transformation function. (c) Equalized histogram.

Histogram equalization is not guaranteed to result in a uniform histogram.

Examples

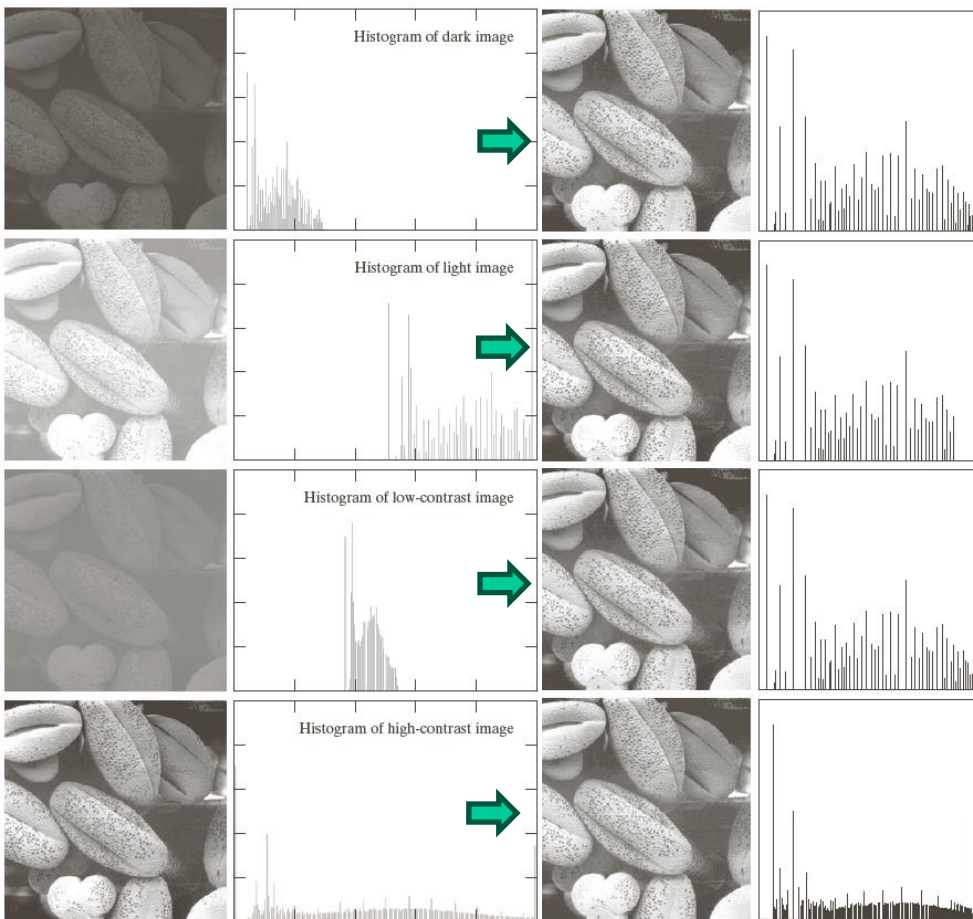
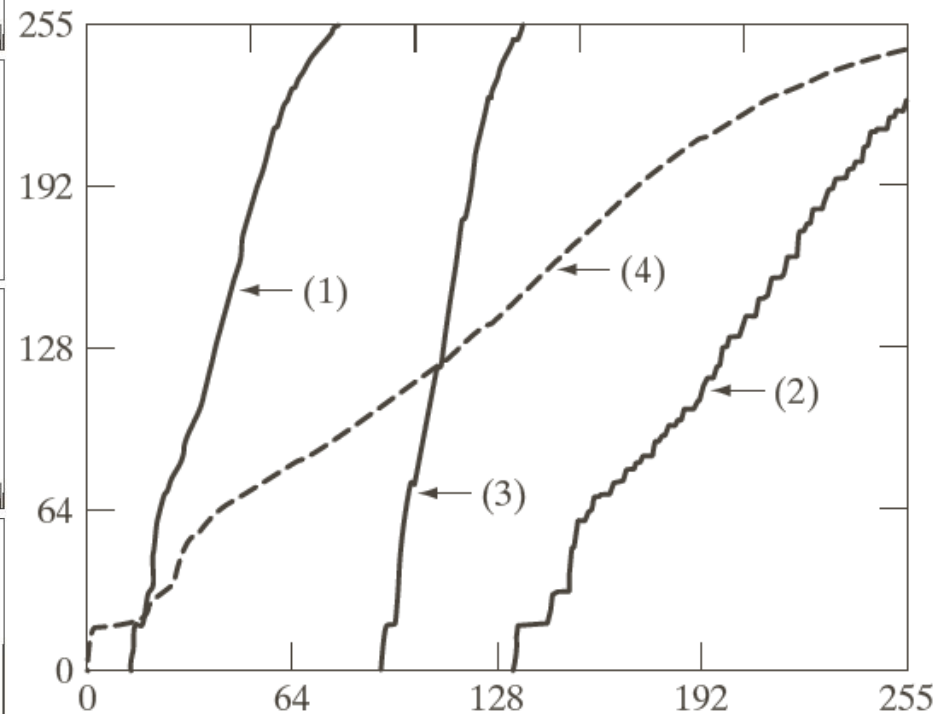
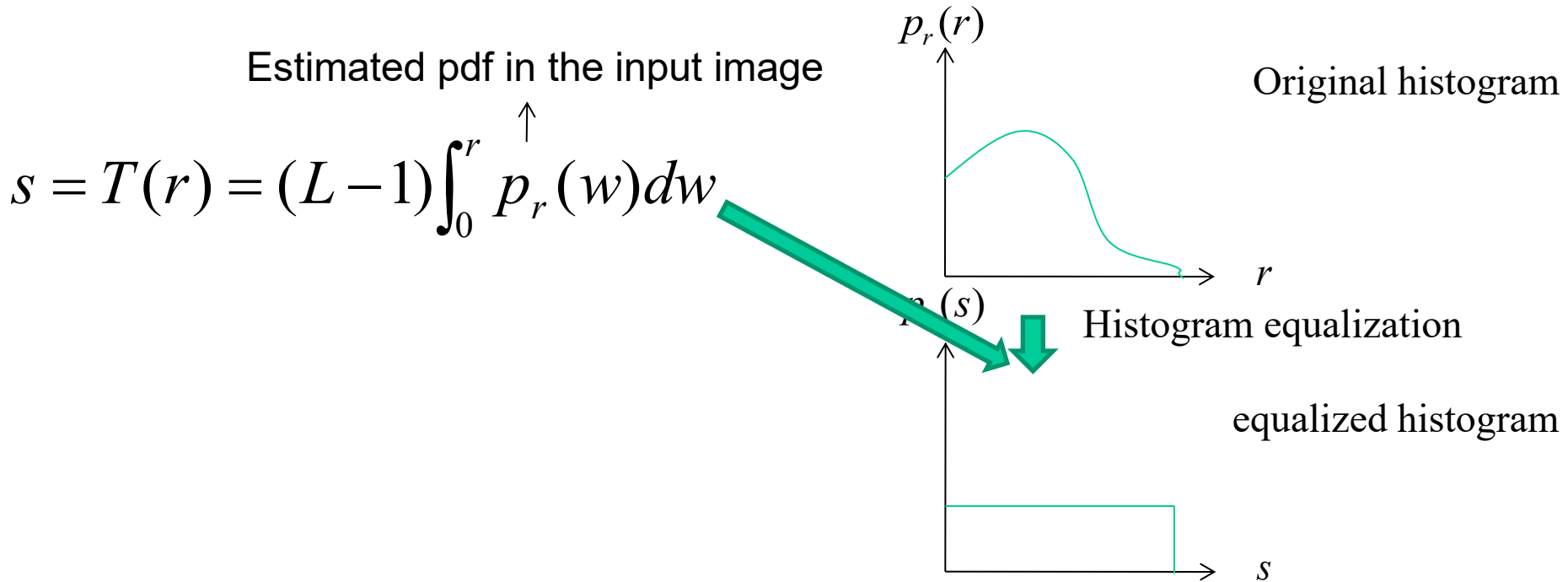


FIGURE 3.20 Left column: images from Fig. 3.16. Center column: corresponding histogram-equalized images. Right column: histograms of the images in the center column.

FIGURE 3.21 Transformation functions for histogram equalization. Transformations (1) through (4) were obtained from the histograms of the images (from top to bottom) in the left column of Fig. 3.20 using Eq. (3.3-8).

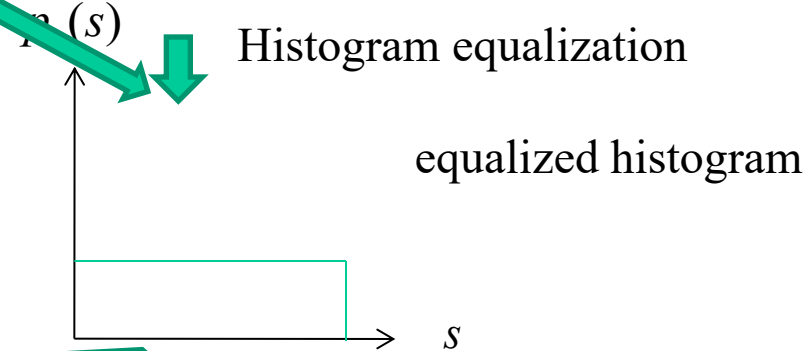
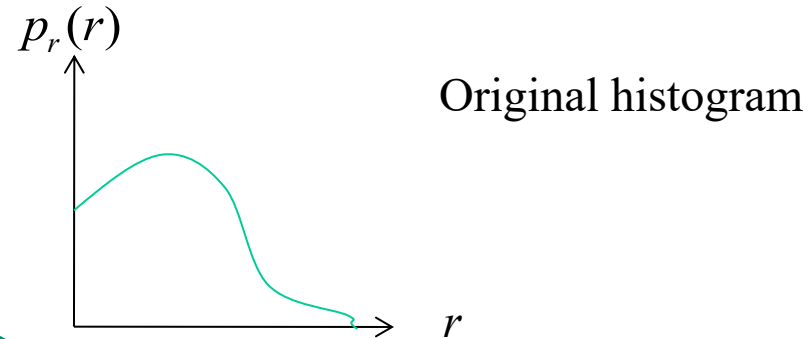


Histogram Matching (Specification)

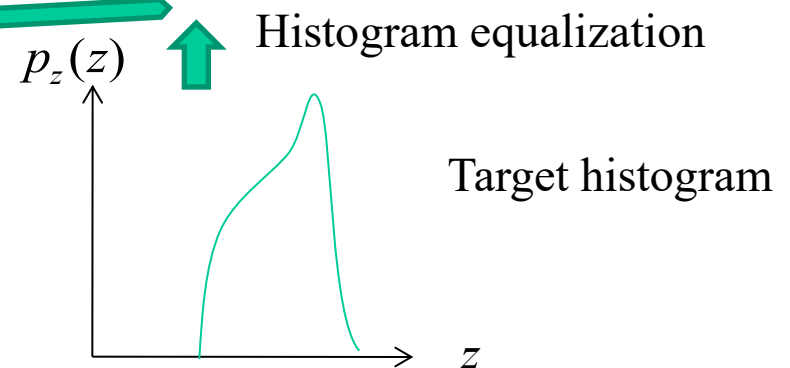


Histogram Matching (Specification)

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$



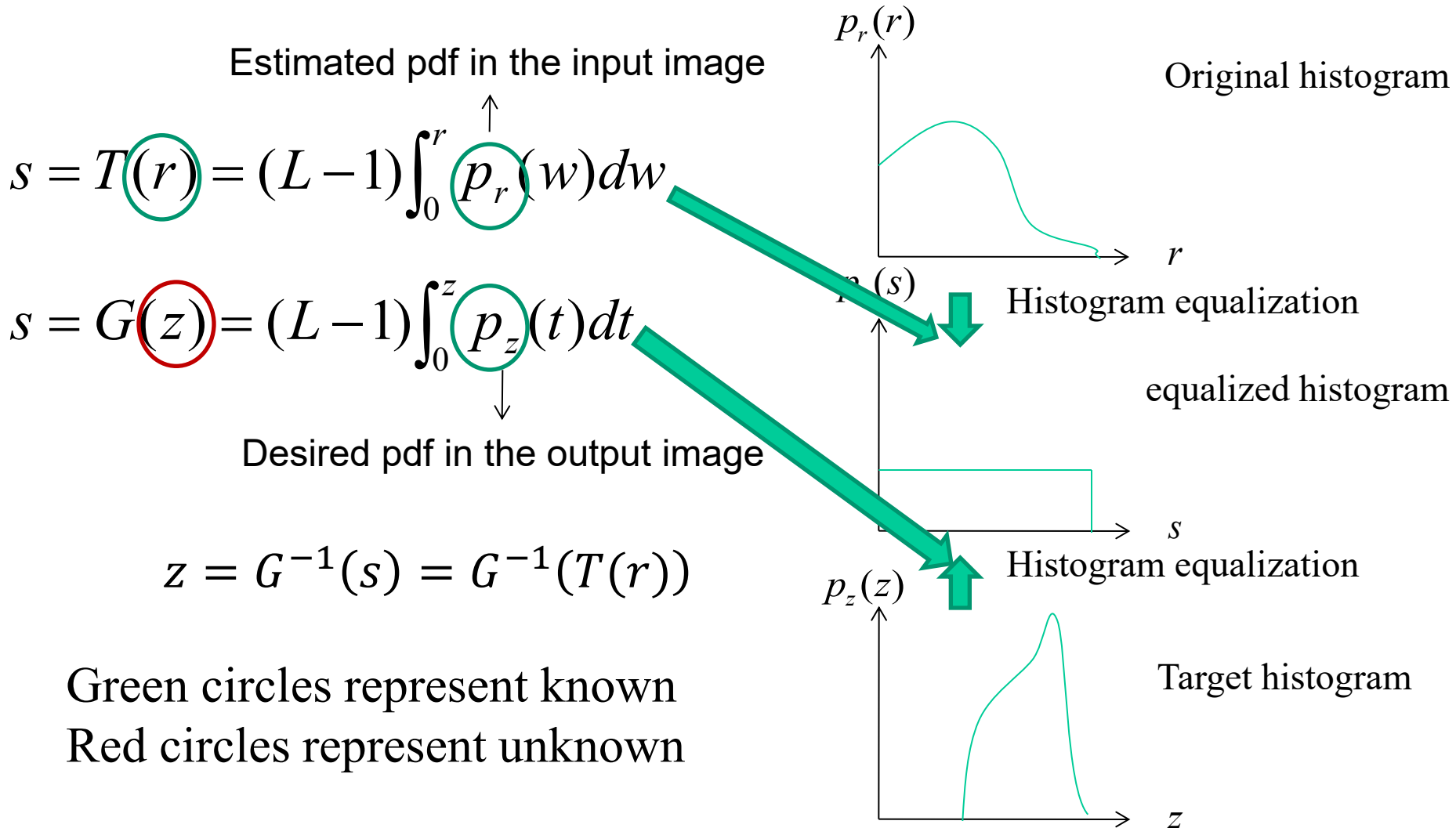
$$s = G(z) = (L-1) \int_0^z p_z(t) dt$$



Histogram equalization

Histogram equalization

Histogram Matching (Specification)



Histogram Matching Algorithm for Continuous Data

Obtain the output image by:

- First compute the probability distribution function of input data $p_r(r)$
- Perform histogram equalization $\rightarrow s = T(r)$
- Compute $s = G(z)$, where G is the equalization function derived from a specified histogram
- Perform the inverse mapping $z = G^{-1}(s) = G^{-1}(T(r))$
- The output image with z values is then of the specified histogram

A Continuous Example

$$p_r(r) = \begin{cases} \frac{2r}{(L-1)^2} & 0 \leq r \leq (L-1) \\ 0 & \textit{otherwise} \end{cases}$$

$$p_z(z) = \begin{cases} \frac{3z^2}{(L-1)^3} & 0 \leq z \leq (L-1) \\ 0 & \textit{otherwise} \end{cases}$$

Compute z ?

Histogram Matching Algorithm – Discrete Image

Discrete histogram require a discretization of the output intensity values

Step1: Compute histogram of the input image $p_r(r)$ and the histogram equalized image $s = T(r)$

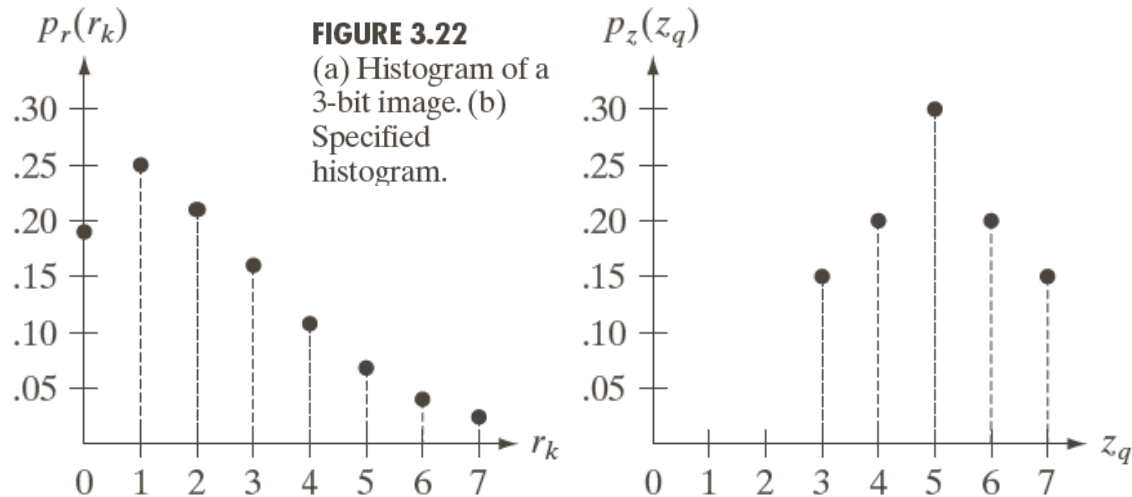
Step2: Compute $G(z)$ given the desired histogram $p_z(z)$

Ideally, $G(z) = s$. In practice, $G(z) \approx s$

Step3: Given the s_k value, find the value of z_q so that $G(z_q)$ is closest to s_k

Step4: form the histogram-specified image using the mapping r-z found above

A Discrete Example



r_k	n_k	$p_r(r_k) = n_k/MN$
$r_0 = 0$	790	0.19
$r_1 = 1$	1023	0.25
$r_2 = 2$	850	0.21
$r_3 = 3$	656	0.16
$r_4 = 4$	329	0.08
$r_5 = 5$	245	0.06
$r_6 = 6$	122	0.03
$r_7 = 7$	81	0.02

z_q	Specified $p_z(z_q)$
$z_0 = 0$	0.00
$z_1 = 1$	0.00
$z_2 = 2$	0.00
$z_3 = 3$	0.15
$z_4 = 4$	0.20
$z_5 = 5$	0.30
$z_6 = 6$	0.20
$z_7 = 7$	0.15

A Discrete Example – Cont.

r_k	n_k	$p_r(r_k) = n_k/MN$	S	G(z)	z
$r_0 = 0$	790	0.19	$S_0=1$	$G(z_0)=0$	$z_0=0$
$r_1 = 1$	1023	0.25	$S_1=3$	$G(z_1)=0$	$z_1=1$
$r_2 = 2$	850	0.21	$S_2=5$	$G(z_2)=0$	$z_2=2$
$r_3 = 3$	656	0.16	$S_3=6$	$G(z_3)=1$	$z_3=3$
$r_4 = 4$	329	0.08	$S_4=6$	$G(z_4)=2$	$z_4=4$
$r_5 = 5$	245	0.06	$S_5=7$	$G(z_5)=5$	$z_5=5$
$r_6 = 6$	122	0.03	$S_6=7$	$G(z_6)=6$	$z_6=6$
$r_7 = 7$	81	0.02	$S_7=7$	$G(z_7)=7$	$z_7=7$

$r_0 \rightarrow z_3$

$r_1 \rightarrow z_4$

$r_2 \rightarrow z_5$

$r_3, r_4 \rightarrow z_6$

$r_5, r_6, r_7 \rightarrow z_7$

z_q	Specified $p_z(z_q)$	Actual $p_z(z_k)$
$z_0 = 0$	0.00	0.00
$z_1 = 1$	0.00	0.00
$z_2 = 2$	0.00	0.00
$z_3 = 3$	0.15	0.19
$z_4 = 4$	0.20	0.25
$z_5 = 5$	0.30	0.21
$z_6 = 6$	0.20	0.24
$z_7 = 7$	0.15	0.11

Histogram Matching Algorithm – Discrete Image

Discrete histogram require a discretization of the output intensity values

Step1: Compute histogram of the input image $p_r(r)$ and the histogram equalized image $s = T(r)$

Step2: Compute $G(z)$ given the desired histogram $p_z(z)$

Ideally, $G(z) = s$. In practice, $G(z) \approx s$

Step3: Given the s_k value, find the value of z_q so that $G(z_q)$ is closest to s_k

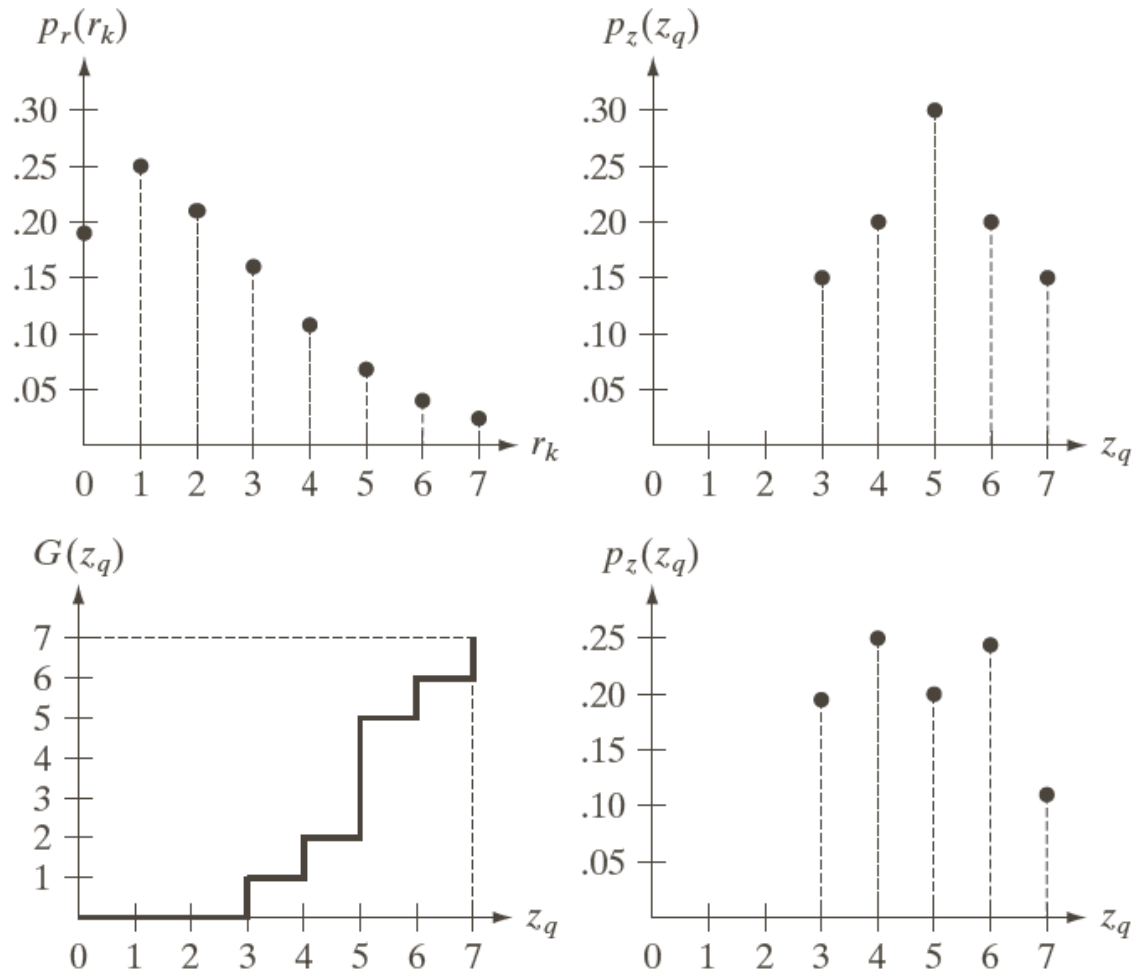
Potential issue: Cause a one-to-multiple mapping

-- multiple z_q are mapped to the same $G(z_q)$

Solution: assign the z-s pair with smallest z_q

Step4: form the histogram-specified image using the mapping r-z found above

An example

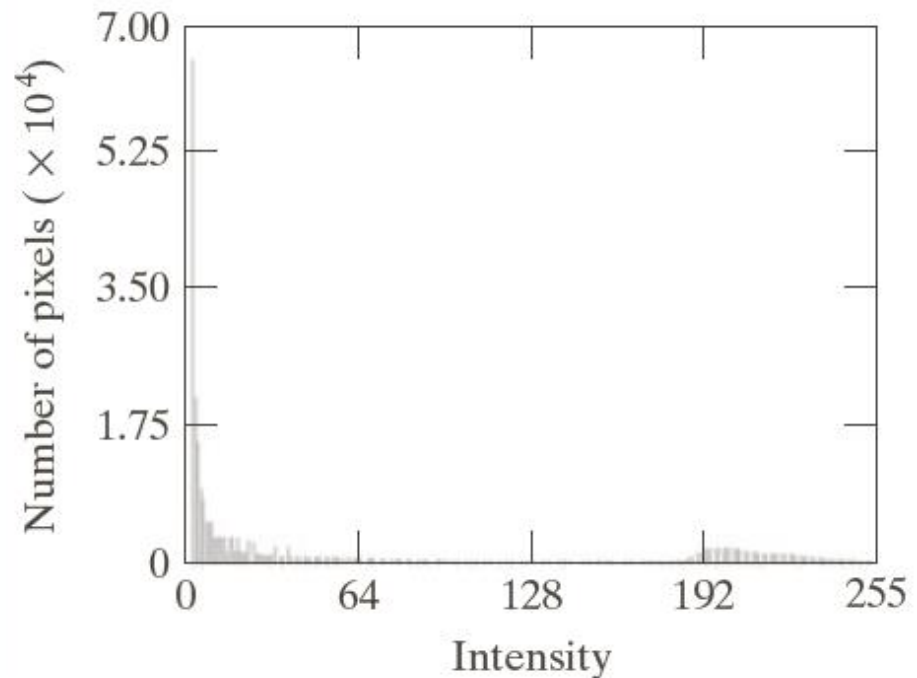


a	b
c	d

FIGURE 3.22

(a) Histogram of a 3-bit image. (b) Specified histogram. (c) Transformation function obtained from the specified histogram. (d) Result of performing histogram specification. Compare (b) and (d).

A Real Example

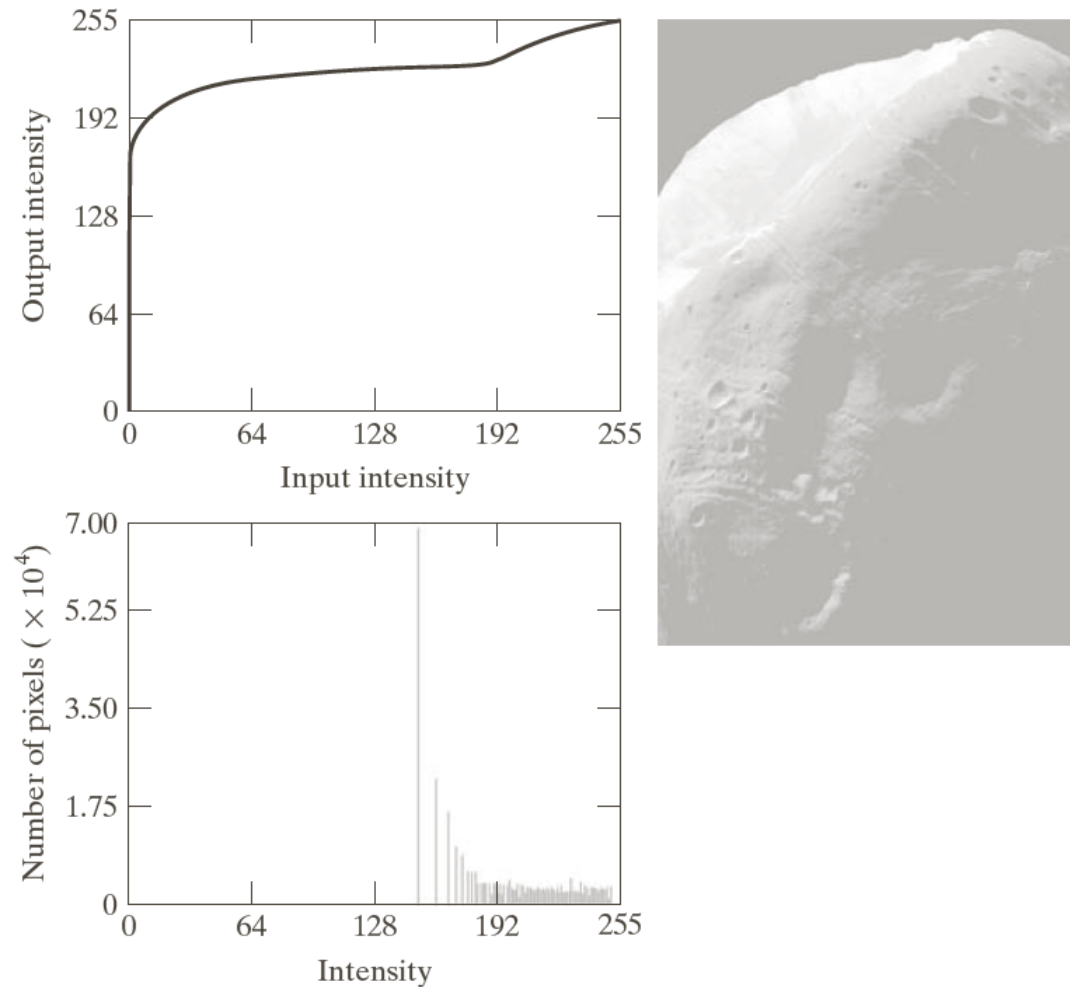


a b

FIGURE 3.23

(a) Image of the Mars moon Phobos taken by NASA's *Mars Global Surveyor*.
(b) Histogram. (Original image courtesy of NASA.)

A Real Example – Histogram Equalization Result



a b
c

FIGURE 3.24

(a) Transformation function for histogram equalization.

(b) Histogram-equalized image (note the washed-out appearance).

(c) Histogram of (b).

A Real Example – Histogram Matching Result

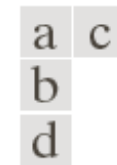
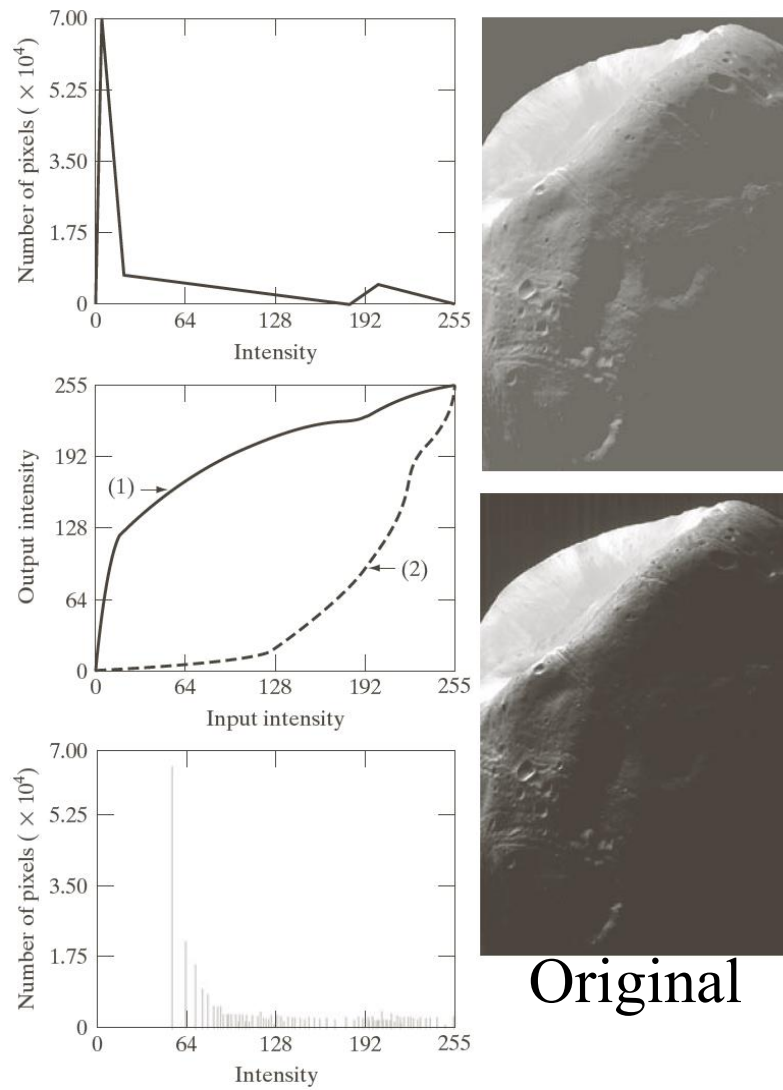


FIGURE 3.25

(a) Specified histogram.
(b) Transformations.
(c) Enhanced image using mappings from curve (2).
(d) Histogram of (c).